Towards a Taxonomy of ICT and Software Sustainability Assessment Instruments

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Abstract: Buyers and regulators of ICT products face instruments that look alike but rest on different methodologies and make substantially different claims. We ask which properties of an instrument determine what it can validly claim, and where deployed instruments fail to realise what the underlying standards support. Following the method of Nickerson et al., we develop in three iterations two coordinated sub-taxonomies, one for the assessment object and one for the sustainability claim, each with five dimensions derived from the ISO 14000 family and from ICT-specific methodologies. Applied to 14 deployed instruments, the taxonomy surfaces a software labelling gap, a structural split between practice-maturity and emission-inventory assessment, a distinction between offset-backed and elimination-based conformance, and the absence of category-specific product category rules for ICT object classes.

Keywords: Sustainability assessment, Software sustainability, ICT environmental labels, Taxonomy development, Software Carbon Intensity

1 Introduction

Buyers, procurers, and regulators of ICT products routinely encounter sustainability instruments (SIs) - e.g. ecolabels, carbon dashboards and other green claims - that are superficially similar but differ in substance. They may all carry a green indicator and a number but rest on different methodologies. Green claims are often misleading and the underlying landscape of frameworks and standards is complex [Mc20], creating a need for a classification that shows what claim can be validly made. Existing reviews collect such instruments but rarely separate *what is being assessed* from *what is being claimed*, so that instruments sharing a label form are treated alike even when they address different objects [Mi20].

Two questions follow: **RQ1**: Which properties of an ICT SI determine which environmental claims it can validly make? **RQ2**: where does the layer of deployed instruments fail to realise claims that the underlying methodologies already support? To answer these questions we propose the development of a taxonomy as a tool to make visible which claims can be validly made about which objects and to navigate the complex field of ICT sustainability assessment as taxonomies are commonly used for this purpose in information systems research [NVM13] and provide an extensible artifact as opposed to capturing a snapshot of this evolving and regulation driven field.

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We use the following terms: A *framework standard* specifies how an assessment is to be carried out, as ISO 14040/14044, ITU-T L.14xx, the Greenhouse Gas (GHG) Protocol ICT Sector Guidance etc. do. An *assessment instrument* is a deployed programme, label, certification, or tool that applies such a methodology and issues a result, and a *claim* is the statement about the assessed object that it makes or enables.

2 Background and Related Work

2.1 Framework standards

The ISO 14000 family, of which ISO 14001 is the most widely adopted member [Ka23], provides the conceptual frame for assessment, reporting, and certification. The ISO 14020 series defines three label types [Mi20]: Type I third-party certified conformance (ISO 14024), Type II self-declared claims (ISO 14021), and Type III quantified ecoprofiles under product category rules (PCR, ISO 14025). ISO 14040/14044 sets out life-cycle assessment principles and the functional-unit construct that enables rate-based accounting, ISO 14064-1 and ISO 14067 quantify emissions at organisational and product level, and ISO 14064-2 covers offset projects. The 2021 amendment to ISO 14021 extended self-declaration to carbon-neutral claims resting either on a zero footprint or on compensating offsets, acknowledging offset-backed substantiation as a distinct path at a time when a substantial literature questions offset integrity [Pr24; SP25].

The ITU-T L.14xx family covers product- and service-level life-cycle assessment (L.1410), organisational GHG and energy accounting (L.1420), and enabling, that is second-order, effects (L.1430), and ETSI ES 203 199 provides a related methodology for ICT goods, networks, and services. The GHG Protocol ICT Sector Guidance adds guidance on how to allocate emissions on shared resources [GH17], while ISO/IEC 21031:2024 specifies the Software Carbon Intensity (SCI) score as a rate of emissions per functional unit of software work, a design that responds to continuously deployed services without a fixed release state [Mo14]. The SDIA Digital Environmental Footprint adds a multi-criteria methodology for digital services that has not moved beyond pilot use.

2.2 Software and sustainability labels

Prior work approaches labelling from two directions. The software-engineering literature established sustainability as a software-quality concern and developed criteria and models for green software [CP15], and ecolabels specific to software have been investigated by Kern et al. [Ke15; Ke18], culminating in the Blue Angel ecolabel for software [NGK20]. The second direction comes from economics and communication studies, which analyse the logic of labelling claims [Bo03] and characterise environmental labels [Mi20].

3 Method

Choice of method and procedure. The taxonomy is developed with the seven-step method of Nickerson et al. [NVM13] which is selected due to its extensibility and the flexibility of offering approaches starting from a conceptual understanding as well as empirical observation. The iterative approach additionally makes it easier to approach a complex field that cannot be digested at a glance. The prior definition of a meta-characteristic provides a clear focus for the taxonomy and the termination criteria offer a mechanism to prevent the taxonomy becoming too fragmented. We retain the original step numbering so that Section 5 can be read against the method: steps 1 and 2 define meta-characteristics and ending conditions; step 3 chooses the approach for the iteration, conceptual-to-empirical (marked *c*) or empirical-to-conceptual (marked *e*); steps 4 to 6 follow that branch, conceiving dimensions (4c), examining objects for them (5c), and forming the revised taxonomy (6c), or selecting objects (4e), identifying common characteristics (5e), and grouping them into dimensions (6e); and step 7 testing the ending conditions.

Two coordinated meta-characteristics. The landscape is governed by four questions: what is assessed, how it is assessed, who asserts and verifies the claim, and for whom it is intended. The properties of the assessed object constrain which claims can validly be made about it and combining it with the assessment method would make it harder to ask which claims an object could support but does not receive, which is necessary to engage with RQ2. The analysis is therefore split into Taxonomy A, "the properties of the assessment object that determine which environmental claims can validly be made about it", and Taxonomy B, "the nature of the claim and the communicative properties of claim that determine its validity, verifiability, and appropriate use context".

The nine objective ending conditions of Nickerson et al. are adopted in full, and the five subjective ones are operationalised as targets: *concise* (7 ± 2 dimensions, 3–5 characteristics each), *robust*, *comprehensive*, *extensible*, and *explanatory* in the sense that at least one finding is surfaced.

4 Instrument Set and Coding

The SIs were identified between January and June 2026 based on prior domain knowledge, documentation of hosting and cloud providers and the standards catalogues of ITU-T, ETSI, ISO and GHG Protocol. An instrument was included if it was deployed and publicly accessible at that time, if its assessment object lies within ICT, and if it issues or directly enables an environmental claim about that object. Excluded were labels proposed only in academic work, general product labels without an ICT-specific object, programmes not yet operating, and the framework standards themselves, which prescribe how to assess rather than issue claims themselves. Each instrument was assigned one characteristic per dimension on the basis of its own primary documentation.

The procedure yields 14 instruments in five strata: the *software ecolabel* Blue Angel DE-UZ 215; the *hosting and data-centre programmes* Green Web Foundation (GWF) Hosting Directory, Green Web Check, the EU Code of Conduct for Data Centres [ABP25], and the Climate Neutral Data Centre Pact (CNDCP); the *SaaS and cloud customer-footprint tools* AWS Sustainability Console [2], Google Cloud Carbon Footprint [3], Microsoft Emissions Impact Dashboard (EID) [4], and Neutraly CO_2Neutral Web Certified (CNWC); the *ICT hardware ecolabels* EPEAT, TCO Certified Gen. 10, and the EU Energy Label for servers; and the *practice-maturity certifications* GreenSeal GSP and Green DiSC [5]. This set is a sample and not an exhaustive list.

5 Iterations

The notes on the full iterations are included in the supplementary material [6]

Iteration 1 (conceptual-to-empirical). The goal is an initial dimension set derived from the framework standards. Step 4c produced five candidate dimensions per sub-taxonomy, among them *Locus of environmental impact* and *Lifecycle stance* on the object side. Step 5c classified four instruments (Blue Angel DE-UZ 215, EPEAT, SCI, and the GHG Protocol ICT Sector Guidance) and surfaced three problems: the first of those two dimensions produced multi-valued cells for Blue Angel, which makes an attributional and an enabling-effects claim at once; the second produced no distinguishing cells and overlapped with scope coverage and claim precision; the audience dimension was multi-valued for Blue Angel. In step 6c the first dimension was split into A3 *Scope coverage* and A4 *Enabling effects included*, the second was dropped as its content is carried by A3 and B2, and the audience conflict with the mutually exclusive and collectively exhaustive (MECE) requirement was deferred to Iteration 2, where the full set provides the evidence for a resolution rule.

Iteration 2 (empirical-to-conceptual). The goal is to test the working taxonomy against all 14 instruments and revise where they do not fit (steps 3 and 4e). Step 5e found two patterns that resist the taxonomy. The practice-maturity certifications GreenSeal GSP and Green DiSC assess no emissions of any artefact but certify organisational capability to engineer for sustainability, so forcing them into A1 *Organisational unit*, a characteristic derived for ISO 14064-1-style inventories, misrepresents the assessed object. The offset-backed claim of Neutraly CNWC fits neither B3 *Self-declared with substantiation*, which expects evidence about the product itself, nor *Third-party verified ecoprofile*, because

[2] https://aws.amazon.com/sustainability/tools/console/
[3] https://docs.cloud.google.com/carbon-footprint/docs/methodology
[4] https://learn.microsoft.com/en-us/power-bi/connect-data/azure-emissions-calculation-methodology
[5] https://www.software.ac.uk/GreenDiSC
[6] https://doi.org/10.5281/zenodo.22211713

its substantiation rests on the integrity of an offsetting scheme. Step 6e therefore adds *Organisational practice/capability maturity* to A1 and *Self-declared with offset substitution* to B3 without altering the dimension structure, and resolves the deferred MECE conflict by a precedence rule for B4: the audience an operator names explicitly takes precedence, and where none is named the form of the instrument decides, so that a public label indicates *Consumer-facing*, a purchasing directory *Procurement/B2B*, an internal dashboard *Engineering/operational*, and so on for the remaining two characteristics. Operators may also exclude an audience, as AWS does when advising against using reported consumption for optimisation[7].

Iteration 3 (conceptual-to-empirical confirmation). The goal is to verify the derived dimensions. A1 *Organisational practice/capability maturity* is the assessment-object counterpart of ISO 14001, which certifies an environmental management system rather than an emission inventory: an organisation can hold ISO 14001 certification without producing an ISO 14064-1 inventory and vice versa. B3 *Self-declared with offset substitution* is anchored in ISO 14021/Amd 1:2021 on the use side and ISO 14064-2 on the supply side. Step 6c adds, merges, and splits nothing further, so all nine objective ending conditions are satisfied, and the subjective ones with two caveats: A1 and B3 carry six characteristics each, one above the preferred range, and several Taxonomy B characteristics have no instance in the set.

6 The Taxonomy

Table 1 states both sub-taxonomies with their characteristics and framework anchors. Three dimensions carry most of the analytical weight and are discussed here; the remainder follow directly from their anchors.

A2 *Substrate coupling* and A5 *Boundary determinacy* carry most of the analytical weight, and they are where software differs systematically from hardware. A2 records how tightly the object is bound to a physical substrate, from hardware that is its own object to software executable across heterogeneous environments whose claims hold only against a declared reference. A5 records whether the object's boundary is unambiguous, dependent on a declared allocation rule as in multi-tenant workloads, conditional on a reference configuration as in substrate-abstracted software, or open at organisational level. It belongs to the object rather than to the methodology: the methodology dictates how a contested boundary is resolved, whereas the object dictates whether it is contested at all [Mo14].

[7] https://docs.aws.amazon.com/sustainability/latest/userguide/energy-calculation.html

Tab. 1: The two sub-taxonomies, their characteristics, and framework anchors. Codes in parentheses are used in Table 2. Characteristics added in Iteration 2 are marked $^{+}$.

Dimension	**Characteristics**	**Anchor**
Taxonomy A — assessment object		
A1 Object granularity	Component (Cp); Discrete artefact (Da); Service or deployment instance (Sd); Organisational unit (Ou); Organisational practice/capability maturity (Pm)$^{+}$; Sector or portfolio (Sp)	ISO 14040/44, 14064-1, 14067, 14001
A2 Substrate coupling	Substrate-identical (Si); Deployment-bound (Db); Substrate-abstracted (Sa); Substrate-aggregated (Sg)	GHG Protocol ICT
A3 Scope coverage	No GHG accounting (–); Operational, Scope 1+2 (Op); Operational + Embodied, Scope 1+2+3 (OE); nested	GHG Protocol scopes
A4 Enabling effects	None (–); Enabling effects claimed (En)	ITU-T L.1430
A5 Boundary determinacy	Determinate (De); Allocation-dependent (Ad); Conditional (Cd); Open or aggregated (Oa)	ISO 14044
Taxonomy B — sustainability claim		
B1 Claim purpose	Conformance (Cf); Performance (Pe); Disclosure (Ds); Pledge or target (Pl)	ISO 14020/24/25, 14064-1
B2 Claim precision	Qualitative (Ql); Ordinal (Or); Absolute quantified (Aq); Rate-based quantified (Rq); Multi-criteria eco-profile (Mq)	ISO 14025, 14067; ISO/IEC 21031
B3 Verification structure	Self-declared with substantiation (Ss); Self-declared with offset substitution (So)$^{+}$; Third-party periodic (Tp); Third-party verified eco-profile (Tv); Independent assurance (Ia); Unverified (Uv)	ISO 14020 types, 14064-3, 14021/Amd 1
B4 Audience and use context	Consumer-facing (Cn); Procurement/B2B (Pr); Engineering/operational (Eg); Regulatory/disclosure (Rg); Investor/ESG (Iv)	ISO 14020
B5 Comparability scope	Non-comparable (Nc); Within programme (Wp); Within PCR or functional-unit class (Wc); Cross-system (Cs); Absolute reference (Ab)	ISO 14025 PCR; functional unit

7 Classification and Gap Analysis

Table 2 classifies all 14 instruments using the codes of Table 1, separating what the standards support from what instruments actually issue.

Tab. 2: Classification of the 14 instruments. One code per dimension; see Table 1 for the code legend.

Instrument	A1	A2	A3	A4	A5	B1	B2	B3	B4	B5
Software ecolabels										
Blue Angel DE-UZ 215	Da	Sa	–	En	Cd	Cf	Ql	Tp	Cn	Wp
Hosting and data-centre programmes										
GWF Hosting Directory	Ou	Sg	Op	–	Oa	Cf	Ql	Ss	Pr	Wp
Green Web Check	Sd	Sg	Op	–	Oa	Cf	Ql	Ss	Cn	Wp
EU CoC for Data Centres	Ou	Si	Op	–	De	Cf	Ql	Ss	Rg	Wp
CNDCP	Ou	Si	Op	–	De	Pl	Aq	Ia	Rg	Wp
SaaS and cloud customer-footprint tools										
AWS Console	Ou	Sg	OE	–	Ad	Ds	Aq	Ia	Rg	Wc
Google Cloud CF	Ou	Sg	OE	–	Ad	Ds	Aq	Ss	Rg	Wc
Microsoft EID	Ou	Sg	OE	–	Ad	Ds	Aq	Ia	Rg	Wc
Neutraly CNWC	Sd	Db	Op	–	Cd	Cf	Ql	So	Cn	Wp
ICT hardware ecolabels										
EPEAT	Da	Si	OE	–	De	Cf	Mq	Tp	Pr	Wp
TCO Certified Gen. 10	Da	Si	OE	–	De	Cf	Mq	Tp	Pr	Wp
EU Energy Label	Da	Si	Op	–	De	Pe	Or	Tp	Pr	Wp
Practice-maturity certifications										
GreenSeal GSP	Pm	Sg	–	–	Oa	Cf	Or	Tp	Eg	Wp
Green DiSC	Pm	Sg	–	–	Oa	Cf	Or	Tp	Eg	Wp

7.1 Labelling-gap register

Three cells are supported by the framework standards but occupied by no instrument. The first is substrate-abstracted software with rate-based quantified disclosure, supported by SCI, which admits self-declared, third-party-verified, and independent-assurance application: no third-party-verified rate-based software disclosure exists. The second is substrate-abstracted software with a multi-criteria ecoprofile, supported by the SDIA methodology and reachable through ETSI ES 203 199 or ITU-T L.1410. The third is deployment-bound objects with *Operational + Embodied* disclosure for individual SaaS or cloud workloads, supported by ETSI ES 203 199, ITU-T L.1410, and SCI. All three describe the same asymmetry: the apparatus to measure software environmental impact exists and, for SCI, is reasonably mature, while the labelling layer that would convert it into market-visible claims is thin.

8 Discussion

Finding 1: Type I conformance labelling for software is rare and absent from service-instance objects. Blue Angel DE-UZ 215 is the only Type I software label in the set, and it works because the programme fixes a reference system against which its criteria are measured [NGK20]. There is no Type I label that covers service-instance software although a programme could specify a reference substrate together with an audit-cycle and certify against them. The existing instruments designed for service-instance objects produce quantified disclosure instead.

Finding 2: The software layer shows a gap between framework support and deployed labelling. All three cells of the labelling-gap register are software-side, so the constraint on software sustainability claims lies in the labelling layer rather than in measurement methodology, which suggests that further methodological refinement is not where the next increment of practical value lies.

Finding 3: Practice-maturity assessment is structurally distinct from emission-inventory assessment. GreenSeal GSP and Green DiSC certify organisational capability to engineer for sustainability, while the vendor customer-footprint tools quantify organisational emissions. Both describe an organisation and are marketed as evidence of organisational sustainability, yet the former is anchored in ISO 14001 on management systems and the latter in ISO 14064-1 on inventories, so they are not interchangeable for compliance reporting such as CSRD and a procurement request should state which kind of evidence it requires.

Finding 4: Offset-backed conformance is structurally distinct from emission-elimination conformance. Neutraly CNWC issues an offset-backed neutrality claim and CNDCP a pledge that uses offsets as part of its pathway, so their substantiation logic differs both from SCI-style elimination and from Type I conformance although all three read identically as “carbon neutral” in marketing, at a time when offset integrity is seriously questioned [Bj22]. The taxonomy reflects the distinction that ISO 14021/Amd 1:2021 recognises by placing Neutraly at *Self-declared with offset substitution* while CNDCP retains *Independent assurance* with offsets recorded as a pledge-pathway annotation. Procurement and regulatory requirements should therefore ask for the substantiation pathway to be declared, not for the neutrality claim alone.

Finding 5: The taxonomy reveals an ICT-specific product category rule gap. ISO 14025 Type III declarations achieve comparability through product category rules that fix scope coverage, life-cycle stages, functional unit, and allocation for one category. The ICT sector has little equivalent infrastructure: ISO/IEC 21031 SCI is a partial analogue for software carbon intensity, the GHG Protocol ICT Sector Guidance is broader, voluntary, and methodology-only. There is no rule set at all for software ecolabels, cloud customer-footprint tools, or practice-maturity certifications. Rules for the principal ICT object classes could be a path towards establishing a way to compare the sustainability of software systems.

9 Limitations

Software-specific ecolabels have a narrow empirical base, with exactly one Type I software label and none for service-instance software, so evidence Findings 1 and 2 is weak. Coding was carried out by the author and no second-coder protocol has been executed, so inter-coder robustness remains unestablished. The set is not exhaustive, excluding emerging instruments such as EPEAT Renew, NADIKI Registrar [8], and those implied by the first EU technical report on data centre energy performance [Di25]. Finally, the taxonomy preserves the MECE requirement on B4 and A3 only through workarounds: B4 relies on a procedural rule that resolves multi-audience instruments at coding time and thereby loses secondary audiences, and A3 uses a nested cumulative scale that cannot represent an instrument covering Scope 3 without Scope 2.

10 Conclusion and Future Work

This paper developed a taxonomy of ICT and software sustainability assessment instruments that separates the assessment object from the sustainability claim. Applied to 14 deployed instruments, RQ1 is engaged by defining ten dimensions that govern claim validity and RQ2 through a register of claims the framework supports but no instrument issues. The central observation is that the ICT sector lacks the category-specific infrastructure that physical-product sustainability has, a gap visible both in the labelling-gap register and in Finding 1, where the only Type I software label solves the problem by fixing a reference substrate at programme level. Future work should include: execution of the second-coder protocol, validation with practitioners in the Green IT community, and engagement with the feasibility of defining product category rules for software.

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